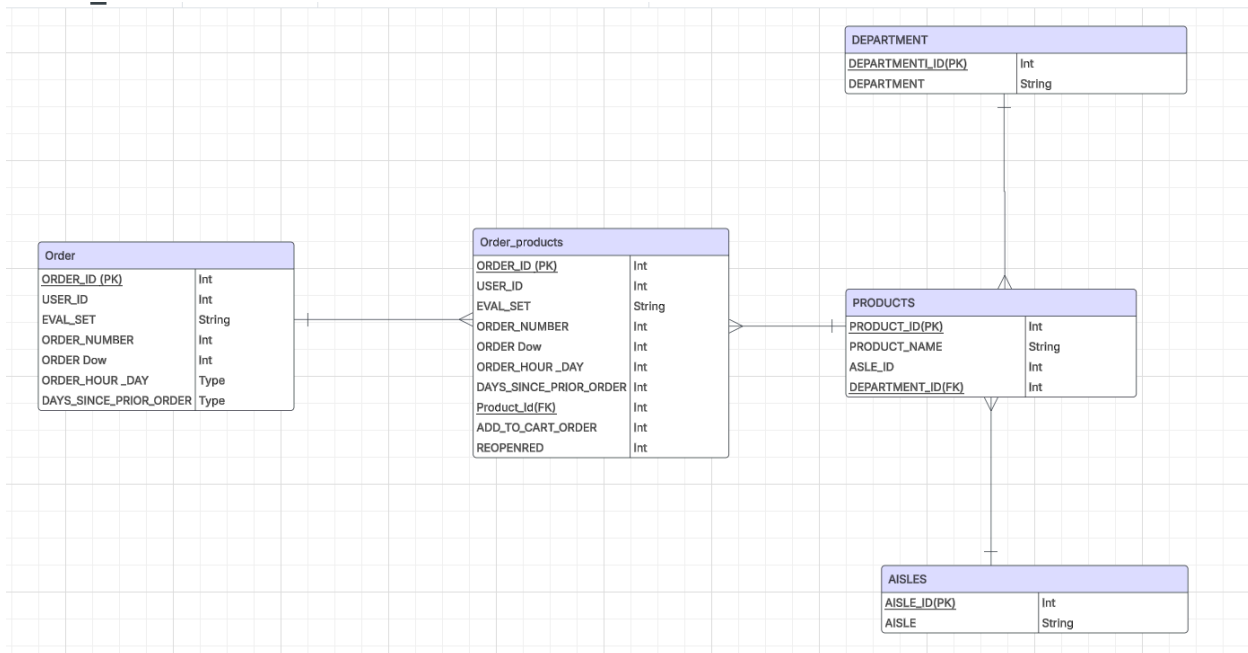


ER Diagram



use ASS1;

#Q1#

```

CREATE TABLE order_products_prior AS
(SELECT a.*,
       b.product_id,
       b.add_to_cart_order,
       b.reordered
FROM   orders a
       JOIN order_products b
       ON a.order_id = b.order_id
WHERE  a.eval_set = 'prior')
    
```

#Q2#

```

select USER_ID, MAX(order_number), AVG(days_since_prior_order) from orders
group by user_id;
    
```

#Q3#

```

select user_id, sum(product_id), sum(distinct product_id), SUM(CASE WHEN REORDERED=1
THEN 1 ELSE 0 END)*1.0/ COUNT(CASE WHEN ORDER_ID>1 THEN 1 ELSE NULL END)
from order_products_prior
group BY user_id;
    
```

#Q4#

```
select * from order_products_prior limit 10;
select user_id, product_id,
count(order_id),
min(order_number),
max(order_number),
avg(add_to_cart_order) from order_products_prior
group by user_id, product_id;
```

#Q5#

WITH allod AS

```
(select user_id, product_id, add_to_cart_order, order_NUMBER, reordered,
ROW_NUMBER() OVER(PARTITION BY user_id, product_id ORDER BY order_NUMBER) as
seq from order_products_prior
)
```

```
select
product_id,
count(*) as total_order,
SUM(CASE WHEN REORDERED=1 THEN 1 ELSE 0 END) as reord,
SUM(CASE WHEN seq=1 THEN 1 ELSE 0 END) as firstbuy,
SUM(CASE WHEN seq=2 THEN 1 ELSE 0 END) as secondbuy
from allod
GROUP BY product_id
ORDER BY product_id;
```